# Session 4 – Data Validation & Cleaning

## Learning Goals

By the end of this session, students should be able to:

- Identify missing values, duplicates, and outliers.
- Handle missing data using imputation or removal.
- Remove duplicate rows.
- Detect and visualize outliers.

### Dataset Setup

Use the following dataset:

```python
import pandas as pd
import numpy as np

data = {
    "Student_ID": [1, 2, 3, 4, 5, 5, 6, 7],
    "Name": ["Alice", "Bob", "Charlie", "David", "Eva", "Eva", "Frank", None],
    "Attendance": [85, None, 78, 92, 88, 88, 95, 80],
    "Assignment_Score": [75, 80, np.nan, 90, 85, 85, 95, 200],
    "Final_Grade": ["B", "A", "C", "A", None, None, "A", "B"]
}

df = pd.DataFrame(data)
print(df)
```

## Checking for Data Issues

Q1. Print the dataset and inspect it.

Q2. Check for missing values in the dataset.

Q3. Display only rows that have missing values.

## Handling Missing Values

Q4. Fill missing Attendance values with the column mean.

Q5. Replace missing Final_Grade with "Unknown".

Q6. Drop rows where Name is missing.

**Handling Duplicates**

Q7. Check if there are duplicate rows.

Q8. Remove duplicate rows.

**Detecting Outliers**

Q9. Describe the dataset statistics.

Q10. Identify outliers using the Interquartile Range (IQR) method.

Q11. Remove outliers from the dataset.

**Visualizing Outliers**

Q12. Plot boxplot to visualize outliers (optional, if matplotlib installed).

**Reflections,**

Q13. What are missing values and how can we handle them?

Q14. Why should we remove duplicates?

Q15. Why is outlier detection important?

Q16. Show what percentage of data is missing in each column.

Q17. Display how many rows were removed during data cleaning.